

Math 520, F15
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score	
1	20		
2	20		
3	20		
4	20		
5	20		
Total	100		

Good Luck !

Problem 1. (20 points) Use the method of **variation of parameters** to find a particular solution $Y(t)$ for the nongomogeneous equation

$$y'' + 4y = \csc 2t, \quad 0 < t < \pi/2.$$

Solution: Characteristic equation:

$$r^2 + 4 = 0$$

$$\Rightarrow r = \pm 2i$$

$$\Rightarrow y_1(t) = \cos 2t, \quad y_2(t) = \sin 2t$$

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2t & \sin 2t \\ -2\sin 2t & 2\cos 2t \end{vmatrix}$$

$$= 2\cos^2 t + 2\sin^2 t = 2$$

$$Y(t) = u_1(t)y_1(t) + u_2(t)y_2(t)$$

$$u_1(t) = \int \frac{-g(t)y_2(t)}{W(y_1, y_2)(t)} dt = - \int \frac{\csc 2t \cdot \sin 2t}{2} dt$$

$$= \int -\frac{1}{2} dt = -\frac{t}{2}$$

$$u_2(t) = \int \frac{g(t)y_1(t)}{W(y_1, y_2)(t)} dt = \int \frac{\csc 2t \cos 2t}{2} dt$$

$$= \int \frac{1}{2} \frac{\cos 2t}{\sin 2t} dt$$

$$= \frac{1}{4} \int \frac{2\cos 2t}{\sin 2t} dt$$

$$= \frac{1}{4} \ln |\sin 2t| = \frac{1}{4} \ln(\sin 2t) \quad \text{Since } 0 < t < \frac{\pi}{2}$$

$$\text{Thus } Y(t) = -\frac{t}{2} \cos 2t + \frac{1}{4} \sin 2t \ln(\sin 2t)$$

Problem 2. (20 points) Given the following nonhomogeneous third-order differential equation

$$y^{(3)} + y' = -\cos t.$$

(a) Find the general solution $y_c(t)$ of the associated homogeneous equation.

Solution: Characteristic equation:

$$r^3 + r = 0$$

$$r(r^2 + 1) = 0$$

$$r_1 = 0, r_2 = i, r_3 = -i$$

$$\Rightarrow y(t) = C_1 + C_2 \cos t + C_3 \sin t$$

(b) Find a particular solution $Y(t)$ for the above equation using the method of **undetermined coefficients**:

$$Y(t) = t^s (A \cos t + B \sin t) \quad \underline{s=1!}$$

$$= At \cos t + Bt \sin t$$

$$Y'(t) = A \cos t - At \sin t + B \sin t + Bt \cos t$$

$$Y''(t) = -A \sin t - A \sin t - At \cos t + B \cos t + B \cos t - Bt \sin t$$

$$= -2A \sin t + 2B \cos t - At \cos t - Bt \sin t$$

$$Y'''(t) = -2A \cos t - 2B \sin t - A \cos t + At \sin t - B \sin t - Bt \cos t$$

$$= -3A \cos t - 3B \sin t + At \sin t - Bt \cos t$$

$$Y^{(3)} + Y' = [-3A \cos t - 3B \sin t + At \sin t - Bt \cos t] + [A \cos t + B \sin t - At \cos t + Bt \sin t]$$

$$= -2A \cos t - 2B \sin t = -\cos t$$

$$\Rightarrow -2A = -1 \Rightarrow A = \frac{1}{2} \Rightarrow Y_H = \frac{1}{2} \cos t$$

$$-2B = 0 \Rightarrow B = 0$$

Problem 3. (20 points)

(a) Find the general solution for the equation: $y^{(6)} - y'' = 0$.

$$\gamma^6 - \gamma^2 = 0$$

$$\gamma^2(\gamma^4 - 1) = 0$$

$$\gamma^2(\gamma^2 - 1)(\gamma^2 + 1) = 0$$

$$\gamma^2(\gamma - 1)(\gamma + 1)(\gamma^2 + 1) = 0$$

$$\Rightarrow \gamma_1 = \gamma_2 = 0, \gamma_3 = 1, \gamma_4 = -1, \gamma_5 = i, \gamma_6 = -i$$

$$\Rightarrow y(t) = C_1 + C_2 t + C_3 e^t + C_4 e^{-t} + C_5 \cos t + C_6 \sin t$$

(b) Find the general solution for the equation: $y^{(4)} - 8y'' = 0$.

$$\gamma^4 - 8\gamma^2 = 0$$

$$\gamma(\gamma^3 - 8) = 0$$

$$\gamma(\gamma - 2)(\gamma^2 + 2\gamma + 4) = 0$$

$$\Rightarrow \gamma_1 = 0, \gamma_2 = 2, \gamma_3 = -1 + \sqrt{3}i, \gamma_4 = -1 - \sqrt{3}i$$

$$\Rightarrow y(t) = C_1 + C_2 e^{2t} + C_3 e^{-t} \cos(\sqrt{3}t) + C_4 e^{-t} \sin(\sqrt{3}t)$$

(c) Find $\mathcal{L}^{-1}\left\{\frac{s+1}{(s-2)^2+9}\right\}$ (Hint: using the formula $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$)

$$\mathcal{L}^{-1}\left\{\frac{s+1}{(s-2)^2+9}\right\} = \mathcal{L}^{-1}\left\{\frac{(s-2)+3}{(s-2)^2+3^2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s-2}{(s-2)^2+3^2}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{(s-2)^2+3^2}\right\}$$

$$= e^{2t} \cos 3t + e^{2t} \sin 3t$$

Problem 4. (20 points) Use Laplace Transform to solve the initial value problem:

$$y'' + 9y = 6t, \quad y(0) = 2, \quad y'(0) = 0.$$

Solution: Let $Y(s) = \mathcal{L}\{y(t)\}$, then

$$\mathcal{L}\{y'' + 9y\} = \mathcal{L}\{6t\}$$

$$\mathcal{L}\{y''\} + 9\mathcal{L}\{y\} = 6\mathcal{L}\{t\}$$

$$[s^2 Y(s) - s y(0) - y'(0)] + [9 Y(s)] = \frac{6}{s^2}$$

$$s^2 Y(s) - 2s + 9 Y(s) = \frac{6}{s^2}$$

$$(s^2 + 9) Y(s) = \frac{6}{s^2} + 2s$$

$$Y(s) = \frac{6}{(s^2 + 9)s^2} + \frac{2s}{s^2 + 9}$$

$$\frac{6}{(s^2 + 9)s^2} = \frac{A}{s^2 + 9} + \frac{B}{s^2} = \frac{As^2 + B(s^2 + 9)}{(s^2 + 9)s^2} = \frac{(A+B)s^2 + 9B}{(s^2 + 9)s^2}$$

$$\Rightarrow A + B = 0 \Rightarrow A = -\frac{2}{3}$$

$$9B = 6 \Rightarrow B = \frac{2}{3}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{6}{(s^2 + 9)s^2} + \frac{2s}{s^2 + 9}\right\}$$

$$= \mathcal{L}^{-1}\left\{\left(-\frac{2}{3}\right)\frac{1}{s^2 + 9} + \left(\frac{2}{3}\right)\frac{1}{s^2} + \frac{2s}{s^2 + 9}\right\}$$

$$= \mathcal{L}^{-1}\left\{-\frac{2}{3}\frac{1}{s^2 + 9}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{3}\frac{1}{s^2}\right\} + \mathcal{L}^{-1}\left\{\frac{2s}{s^2 + 9}\right\}$$

$$= \left(-\frac{2}{3}\right) \cdot \frac{1}{3} \sin 3t + \frac{2}{3}t + 2\cos 3t$$

$$= -\frac{2}{9} \sin 3t + \frac{2}{3}t + 2\cos 3t$$

Problem 5. (20 points) Find the general solution of the following homogeneous system of differential equations:

$$\frac{dx}{dt} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & -3 \\ 0 & -2 & 1 \end{pmatrix} x.$$

Solution: $A = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & -3 \\ 0 & -2 & 1 \end{pmatrix}$

$$\det(A - rI) = \begin{vmatrix} 1-r & 0 & 3 \\ 0 & 2-r & -3 \\ 0 & -2 & 1-r \end{vmatrix} = (1-r)(2-r)(1-r) - 6$$

$$= (1-r)(2-3r+r^2-6)$$

$$= (1-r)(r^2-3r-4)$$

$$= (1-r)(r+1)(r-4)$$

$$\Rightarrow r_1 = 1, r_2 = -1, r_3 = 4.$$

For $r_1 = 1$, we solve $(A - r_1 I)\mathbf{f} = 0$

$$\begin{pmatrix} 0 & 0 & 3 \\ 0 & 1 & -3 \\ 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = 0, \text{ let } f_1 = 1.$$

$$\Rightarrow \mathbf{f}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t$$

For $r_2 = -1$, we solve $(A - r_2 I)\mathbf{f} = 0$

$$\begin{pmatrix} 2 & 0 & 3 \\ 0 & 3 & -3 \\ 0 & -2 & 2 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = 0$$

$$\Rightarrow 2f_1 + 3f_3 = 0$$

$$3f_2 - 3f_3 = 0$$

let $f_3 = 1$, we get

$$\mathbf{f}^{(2)} = \begin{pmatrix} -\frac{2}{3} \\ 1 \\ 1 \end{pmatrix} \Rightarrow \mathbf{x}^{(2)}(t) = \begin{pmatrix} -\frac{2}{3} \\ 1 \\ 1 \end{pmatrix} e^{-t}$$

For $r_3 = 4$, we solve $(A - r_3 I)\mathbf{f} = 0$

$$\begin{pmatrix} -3 & 0 & 3 \\ 0 & -2 & -3 \\ 0 & -2 & -3 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = 0$$

$$-3f_1 + 3f_3 = 0$$

$$-2f_2 - 3f_3 = 0$$

$$\text{let } f_3 = 1 \Rightarrow$$

$$\mathbf{f}^{(3)} = \begin{pmatrix} 1 \\ -\frac{3}{2} \\ 1 \end{pmatrix}$$

$$\Rightarrow \mathbf{x}^{(3)}(t) = \begin{pmatrix} 1 \\ -\frac{3}{2} \\ 1 \end{pmatrix} e^{4t}$$

Thus

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} -\frac{2}{3} \\ 1 \\ 1 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} 1 \\ -\frac{3}{2} \\ 1 \end{pmatrix} e^{4t}$$